Q2 - The Midnight Episode (ECG Arrhythmia)

Catching the Arrhythmia via normalized template correlation

Mayur - Roll 240643 - EE200 Signals, Systems & Networks - IIT Kanpur

Real Holter data: patient_ecg.npy (N = 5000 samples) and template.npy (L = 200 samples), fs = 250 Hz. Every number below is computed in the notebook, not eyeballed.

(a) Reading the signal

(i) duration = N/fs = 5000/250 = 20.0 s.
(ii) heart rate = 60/0.8 = 75 bpm; samples/beat = 0.8-250 = 200.
(iii) fundamental f0 = 1/0.8 = 1.25 Hz.

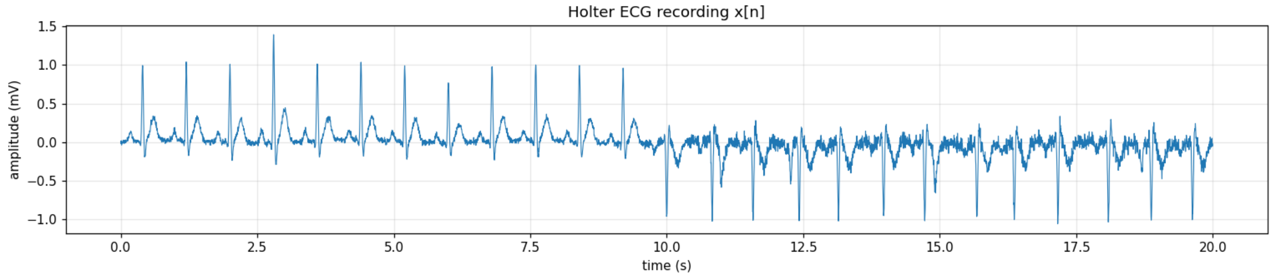

Fig 1. The 20 s recording: regular beats then an irregular region.

(b) Healthy heart in the frequency domain

(i) A (nearly) periodic signal has a DISCRETE LINE spectrum (harmonics of f0), not a smooth continuous curve.
(ii) The sharp, narrow QRS spike drives the HIGH-frequency content (narrow in time => broadband in frequency, time-bandwidth principle); the broad P and T waves are inherently low-frequency.
(iii) At 150 bpm the period halves to 0.4 s, so f0 = 2.5 Hz and the harmonic spacing (= f0) DOUBLES.

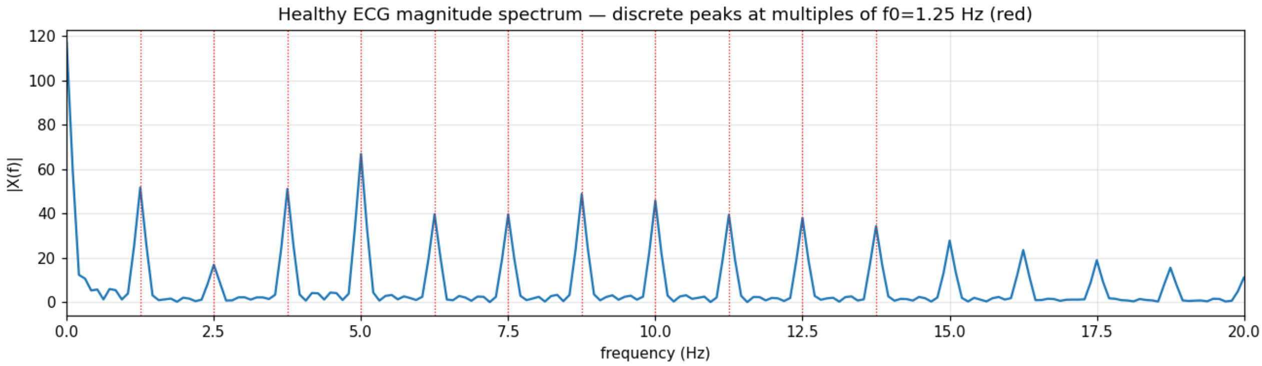

Fig 2. Healthy-segment spectrum: peaks at multiples of f0 = 1.25 Hz (red).

(c) Cutting a heartbeat (windowing)

(i) Window width ~= 200 samples (one beat period), placed to bracket a full {P,QRS,T} starting just before the P wave.
(ii) 80 samples = too short: slices into the beat, clips QRS/T, and the hard rectangular edges cause spectral leakage. 600 samples = too long: captures fragments of neighbouring beats, contaminating the template.
(iii) Same uncertainty trade-off as STFT windows: shorter = sharper time but poorer frequency resolution and here it risks cutting the beat - so 'as short as possible' is not automatically best.

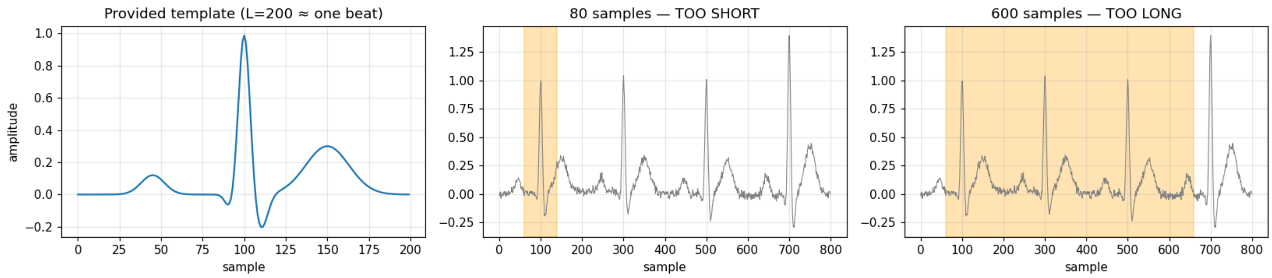

Fig 3. Template and the two bad window widths.

(d) Matching the template (normalized correlation)

rho(m) = Sum t[k]x[m+k] / (||t||||x_m||).

(i) rho in [-1, 1] (Cauchy-Schwarz); +1 = perfect match.

(ii) Normalization removes amplitude/baseline-wander sensitivity: a beat twice as tall but identically shaped keeps rho = +1, while the unnormalized dot product would roughly double - wrongly conflating shape with size.

(iii) An inverted beat gives rho ~= -1, far from the healthy +1, so it is trivially flagged by a threshold.

(e) Onset detection & the spectrogram

(i) Rule: first beat with rho < threshold (0.5). Too high => false alarms from normal variability; too low => misses mild morphology changes.

(ii) Spectrogram: healthy region shows clean steady harmonic bands; the arrhythmia region shows smeared/broken/shifting bands and new irregular energy.

(iii) They can disagree because the spectrogram must accumulate several irregular beats in its window before structure visibly degrades (delayed, smeared), whereas correlation flags per beat - so trust CORRELATION for the exact onset, spectrogram for corroboration.

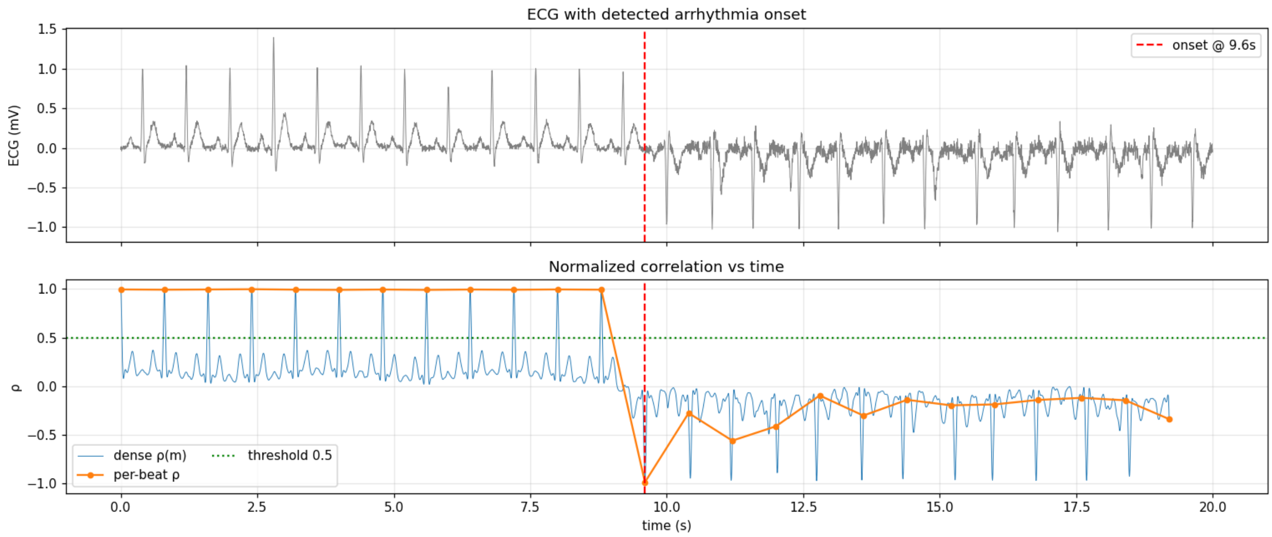

Fig 4. ECG with detected onset (top) and rho vs time (bottom). The first 12 beats score ~= +1; the arrhythmia onset at t = 9.6 s (sample 2400) begins with an inverted beat (rho ~= -1).

(f) Sampling & aliasing

(i) Nyquist minimum = 2-40 = 80 Hz.

(ii) At 50 Hz (< 80 Hz) the QRS content above 25 Hz aliases down, distorting the sharp spike - dangerous because the detector relies on shape correlation, so aliasing causes false positives/negatives.

(iii) Fix: anti-alias low-pass (~25 Hz cutoff) before downsampling; unavoidable cost = losing genuine fine QRS diagnostic detail.

(g) The detector - find_onset

find_onset(ecg, template, threshold) steps beat-by-beat (stride = len(template) = 200), scores each beat by normalized correlation, and returns the first index with rho < threshold (else -1). On the provided data with threshold 0.5 it returns sample 2400 (t = 9.6 s). Implementation in src/q2_ecg.py.

(h) Spectrogram window length

Frequency resolution Deltaf = fs/nperseg. To resolve the f0 = 1.25 Hz harmonic spacing we need Deltaf <~ 1.25 Hz => nperseg >~ 250/1.25 = 200. We use nperseg = 500 (Deltaf = 0.5 Hz) for clean, well-separated harmonic bands.

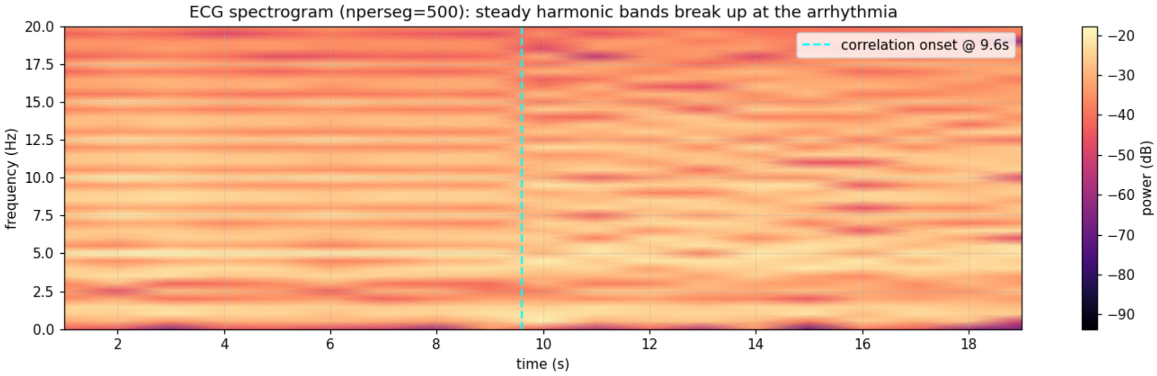

Fig 5. Spectrogram (nperseg = 500): steady harmonic bands break up at the arrhythmia onset (cyan line), corroborating the correlation result.